1 Edge–face product blocks for material-dependent star-local updates

For a homogeneous material, every term of the star-local curl-curl update carries the same weight. A single star-oracle, combined with the cyclic permutations of the component labels, then produces the full update. With a spatially varying reluctivity $\nu = 1/\mu$, each of the twelve neighbouring edges in the star of an edge is weighted by the Hodge coefficient of the face it shares with that edge. This section shows that the material-dependent update can still be written entirely in terms of *index translations* of a small, fixed set of edge-aligned product arrays. These translations are the operations that are cheap on a quantum register.

1.1 Face families

The edge cochain A is mapped to face fluxes by the incidence matrix $\mathbf{D}$, and the Hodge operator $\mathbf{H}$ is diagonal on the faces, so that $M = \mathbf{D}^{\mathsf{T}}\mathbf{H}\mathbf{D}$. On a regular hexahedral complex we group the faces by their normal direction $c \in \{x, y, z\}$. The dual edge of a face crosses the two cells adjacent to it, so on a uniform grid the Hodge coefficient is, up to a common geometric factor, the average of the two cell reluctivities:

$$F_c[p] \;=\; \tfrac{1}{2}\big(\nu_{\text{cell}}[p - \mathbf{e}_c] + \nu_{\text{cell}}[p]\big), \qquad c \in \{x, y, z\}. \tag{1}$$

On non-uniform or deformed grids, (1) is replaced by the length-weighted expression $(\nu_1\ell_1 + \nu_2\ell_2)/|\omega_f|$. Nothing below depends on this choice. The three arrays F_x, F_y, F_z hold the complete material and metric information, and each face is stored exactly once.

1.2 Star-local update in frame coordinates

As in the homogeneous case, the update of each edge component is written in the frame of that component. Consider first the z-component. All arrays are indexed as $[k, i, j]$ with $(k, i, j) = (z, x, y)$, so that $X[k, i, j]$ refers to the physical position $(x, y, z) = (i, j, k)$. Then

- $F_y[k, i, j]$ is the face normal to y at $y = j$, spanning $x \in [i, i{+}1]$ and $z \in [k, k{+}1]$; it lies between the cells $[k, i, j{-}1]$ and $[k, i, j]$;
- $F_x[k, i, j]$ is the face normal to x at $x = i$, spanning $y \in [j, j{+}1]$ and $z \in [k, k{+}1]$; it lies between the cells $[k, i{-}1, j]$ and $[k, i, j]$.

The edge $A_z[k, i, j]$ lies on four faces: two of family F_y (east and west) and two of family F_x (north and south). They contribute

$$\begin{aligned}
E &= F_y[k, i, j]\,\big(A_z[k, i{+}1, j] + A_x[k, i, j] - A_x[k{+}1, i, j]\big),\\
W &= F_y[k, i{-}1, j]\,\big(A_z[k, i{-}1, j] - A_x[k, i{-}1, j] + A_x[k{+}1, i{-}1, j]\big),\\
N &= F_x[k, i, j]\,\big(A_z[k, i, j{+}1] + A_y[k, i, j] - A_y[k{+}1, i, j]\big),\\
S &= F_x[k, i, j{-}1]\,\big(A_z[k, i, j{-}1] - A_y[k, i, j{-}1] + A_y[k{+}1, i, j{-}1]\big),
\end{aligned} \tag{2}$$

and the damped Jacobi step reads

$$\begin{aligned}
A_z^{s+1}[k, i, j] &= \frac{\beta}{\Delta}\big(E + W + N + S + J_z[k, i, j]\big) + (1 - \beta)\, A_z^{s}[k, i, j],\\
\Delta &= F_y[k, i, j] + F_y[k, i{-}1, j] + F_x[k, i, j] + F_x[k, i, j{-}1].
\end{aligned} \tag{3}$$

The y- and x-components obey the same equations in their own frames, obtained by the cyclic relabelling $x \to y \to z \to x$ (Table 1). To treat all frames at once, we write a for the updated component, b and c for the two transverse components, Φ for the east/west face family (normal to the local j-axis) and Ψ for the north/south face family (normal to the local i-axis). Equations (2)–(3) then hold in every frame with $(A_z, A_x, A_y, F_y, F_x, J_z)$ replaced by (a, b, c, Φ, Ψ, J). For the z-frame, $\Phi = F_y$ and $\Psi = F_x$.

frame	(k,i,j)	a	b	c	Φ (E/W)	Ψ (N/S)
z	(z,x,y)	A_z	A_x	A_y	F_y	F_x
y	(y,z,x)	A_y	A_z	A_x	F_x	F_z
x	(x,y,z)	A_x	A_y	A_z	F_z	F_y

Table 1: Frame-local identification of edge components and face families. Each frame is obtained from the previous one by the cyclic relabelling $x \to y \to z \to x$, which acts in the same way on edge components and on face families.

1.3 Why products must be formed per face slot

Let $\mathcal{T}_\delta$ denote the translation $(\mathcal{T}_\delta X)[p] = X[p+\delta]$. A translation acts on a product as a whole:

$$\mathcal{T}_\delta\,(F \circ A) = (\mathcal{T}_\delta F) \circ (\mathcal{T}_\delta A), \tag{4}$$

so the face coefficient and the edge value are always transported together. For example,

$$\mathcal{T}_{+\mathbf{e}_j}(F_x \circ A_z)[k,i,j] = F_x[k,i,j{+}1]\,A_z[k,i,j{+}1],$$

whereas the north term in (2) requires $F_x[k,i,j]\,A_z[k,i,j{+}1]$. Once a coefficient and a field value are combined into one amplitude, no permutation and no linear combination of amplitudes can separate them again.

The way around this is to multiply each edge by *each of its own incident faces*, before any translation is applied. A translation then carries the correct shared face along with the neighbouring edge.

1.4 Edge-aligned product blocks

For each edge, its two faces of each relevant family are labelled by a slot $\sigma \in \{0,-\}$: the face at the same index (0), or the face one step back along the third direction ($-$). This gives eight blocks, each with the shape of its edge array:

$$\begin{aligned}
Q_a^{\Phi 0}[k,i,j] &= \Phi[k,i,j]\,a[k,i,j], & Q_a^{\Phi -}[k,i,j] &= \Phi[k,i{-}1,j]\,a[k,i,j],\\
Q_a^{\Psi 0}[k,i,j] &= \Psi[k,i,j]\,a[k,i,j], & Q_a^{\Psi -}[k,i,j] &= \Psi[k,i,j{-}1]\,a[k,i,j],\\
Q_b^{\Phi 0}[k,i,j] &= \Phi[k,i,j]\,b[k,i,j], & Q_b^{\Phi -}[k,i,j] &= \Phi[k{-}1,i,j]\,b[k,i,j],\\
Q_c^{\Psi 0}[k,i,j] &= \Psi[k,i,j]\,c[k,i,j], & Q_c^{\Psi -}[k,i,j] &= \Psi[k{-}1,i,j]\,c[k,i,j].
\end{aligned} \tag{5}$$

Entries whose face lies outside the complex are zero. Over the three frames these are the $3 \times 4 = 12$ products of each edge with its four incident faces, and each frame uses eight of them.

Proposition. The star contributions (2) are sums of translated blocks:

$$\begin{aligned}
E &= \mathcal{T}_{+\mathbf{e}_i} Q_a^{\Phi -} \;+\; Q_b^{\Phi 0} \;-\; \mathcal{T}_{+\mathbf{e}_k} Q_b^{\Phi -},\\
W &= \mathcal{T}_{-\mathbf{e}_i} Q_a^{\Phi 0} \;-\; \mathcal{T}_{-\mathbf{e}_i} Q_b^{\Phi 0} \;+\; \mathcal{T}_{+\mathbf{e}_k - \mathbf{e}_i} Q_b^{\Phi -},\\
N &= \mathcal{T}_{+\mathbf{e}_j} Q_a^{\Psi -} \;+\; Q_c^{\Psi 0} \;-\; \mathcal{T}_{+\mathbf{e}_k} Q_c^{\Psi -},\\
S &= \mathcal{T}_{-\mathbf{e}_j} Q_a^{\Psi 0} \;-\; \mathcal{T}_{-\mathbf{e}_j} Q_c^{\Psi 0} \;+\; \mathcal{T}_{+\mathbf{e}_k - \mathbf{e}_j} Q_c^{\Psi -},
\end{aligned} \tag{6}$$

where all terms are evaluated at $[k,i,j]$.

Proof. Each identity follows from (5). For example, $\mathcal{T}_{+\mathbf{e}_i} Q_a^{\Phi -}[k,i,j] = \Phi[k,i,j]\,a[k,i{+}1,j]$ and $\mathcal{T}_{+\mathbf{e}_k} Q_b^{\Phi -}[k,i,j] = \Phi[k,i,j]\,b[k{+}1,i,j]$, which are the first and last terms of E. □

The rule is simple. A slot-0 block is read at offset 0 or $-\mathbf{e}$, and a slot-$-$ block is read at offset $+\mathbf{e}$. The offsets that occur are

$$\delta \in \{0,\ \pm\mathbf{e}_i,\ \pm\mathbf{e}_j,\ +\mathbf{e}_k,\ +\mathbf{e}_k - \mathbf{e}_i,\ +\mathbf{e}_k - \mathbf{e}_j\},$$

so every term needs at most two unit translations, along different axes.

1.5 Cyclic symmetry

The face families carry a direction label (the face normal), so they transform under the cyclic rotation π in exactly the same way as the edge components:

$$\pi: \quad (A_x, A_y, A_z) \mapsto (A_y, A_z, A_x), \qquad (F_x, F_y, F_z) \mapsto (F_y, F_z, F_x),$$

together with the rotation of the index axes $(z, x, y) \mapsto (y, z, x)$. After applying π^s, the canonical slots (a, b, c, Φ, Ψ) hold the data of frame s, and the same stencil (6) updates all three components.

By contrast, the per-edge coefficients $\Phi[k, i, j], \Phi[k, i-1, j], \ldots$ do *not* transform this way. They are frame-dependent selections of two face families, and they agree across frames only if ν itself is invariant under π.

1.6 Quantum circuit implementation

The quantum state holds only the fields and sources,

$$\Lambda = [\, A_x, A_y, A_z, J_z, J_y, J_x \,],$$

and each block is padded *per axis* to 2^q entries. The index register then splits into three q-qubit groups $|p_k\rangle|p_i\rangle|p_j\rangle$. A unit translation $\mathcal{T}_{\pm\mathbf{e}}$ is an incrementer or decrementer acting on a single group, with no carries between axes, and the frame rotation π is a relabelling of qubit groups. One relaxation step for the canonical frame is implemented as follows.

1. **Face-slot superposition.** Hadamard gates on a slot register select one of the incident faces of each edge.

2. **Hodge oracle.** An oracle O_H, diagonal in (edge index, slot), multiplies each amplitude by the coefficient of the selected face. The face index is p or $p - \mathbf{e}_t$ and is computed from p in place, so no face register is required. For piecewise-constant materials, O_H consists of two cell-membership tests (comparators on the coordinate registers) followed by a controlled R_Y rotation by $2\arcsin(F/F_{\max})$ on an ancilla. After this step the register holds the blocks (5).

3. **Translations.** A term register selects the twelve terms of (6). Each term applies at most two controlled unit translations to the index groups, so its gate cost is $O(q)$.

4. **Signs and summation.** The orientation signs ± 1 are applied as controlled phases, and the interference-based summation combines the branches into $E+W+N+S$. The diagonal Δ is obtained from O_H by summing the four face slots, and enters the relaxation weight.

5. **Frames.** The y- and x-components are updated by conjugating the same circuit with π and π^2. On the Hodge oracle this conjugation shifts the family label by $-s \pmod 3$ and rotates the index back to physical coordinates, so the material map is always evaluated in physical space.

The cyclic incrementers wrap around at index $2^q - 1$, and the wrapped amplitudes land on boundary edges, which are not updated. The zero planes introduced by the per-axis padding prevent contamination of interior terms.

The construction keeps the separation between combinatorial and metric data. Translations and signs (the star-oracle) are independent of the material, and all material and metric information enters through the single, frame-independent Hodge oracle on F_x, F_y, F_z.

1.7 Numerical verification

We implemented the block formulation (5)–(6) classically, with each translation realised as a shifted array read, and compared it with a direct loop implementation of (2)–(3).

- **Single step.** For the L-shaped object of relative permeability $\mu_r = 10$, on a 16^3 grid with random fields and sources, the block update agrees with the direct update to within 1.8×10^{-15} in all three components.
- **Full solve.** With the boundary conditions of the benchmark problem, both implementations converge in the same number of iterations (365) to identical solutions.
- **Shift table.** All 3×12 entries of the shift table were checked individually on an 8^3 grid, using an L-shaped block of finite height ($\mu_r = 100$) that has no rotational symmetry.
- **Counter-examples.** Translating the products in the opposite slots, as in the counter-example below (4), reproduces the result inside homogeneous regions but produces $O(1)$ errors at material interfaces. This confirms that the slot assignment in (5) is essential.